Hidden Markov Model Gaussian Mixture Volatility Regime Switching Filter: Real-Time Bayesian Forward Probabilistic Conditioning and Capital Preservation

Berry Wahlberg
Berrytrade Quantitative Research, Singapore
research@berrytrade.org

October 2026

Abstract

This research monograph provides an institutional-grade empirical evaluation of the **HMMRegimeEA** trading strategy under the standardized Berrytrade MetaTrader 5 quantitative benchmark. Evaluated across a continuous three-year horizon (January 1, 2023, to December 31, 2025) with tick-level granularity, realistic broker commissions ($7.00/lot), dynamic interbank floating spreads, and 120 ms execution latency, we scrutinize both the mathematical formulation and empirical microstructure performance of the algorithm. Our findings demonstrate that while marketing backtests report a net profit of 15,480 USD with a nominal profit factor of 1.74, the underlying risk architecture exposes capital to extreme left-tail fragility. We document maximum relative equity drawdowns reaching **8.90%** and account margin levels collapsing to **890.0%**. Through statistical moment decomposition, Monte Carlo permutation testing (1,000 reshuffled paths), and microstructure trade duration diagnostics, we separate genuine predictive alpha from the deceptive facade of unhedged loss-averaging overlays.

1 Introduction and Theoretical Motivation

Algorithmic trading in retail foreign exchange, equity indices, and precious metals markets has experienced unprecedented growth, fueled by automated platforms such as MetaTrader 5 (MT5) and accessible MQL5 scripting environments. However, the academic literature in empirical market microstructure and asset pricing consistently highlights a profound disconnect between optimistic backtest claims and dismal real-world survival rates [10, 17, 19, 4, 13]. While institutional quantitative asset managers operate under strict statistical validation frameworks [18, 3], retail automated trading largely relies on simplistic indicator combinations and heuristics prone to extreme selection bias and data mining.

In this monograph, we conduct a rigorous, execution-constrained evaluation of the **HMMRegimeEA** Expert Advisor, archived in public open-source quantitative repositories, benchmarking it against institutional standards established in the Berrytrade Quantitative Research framework [23]. Our investigation is guided by two fundamental quantitative questions:

1. Does the underlying technical signal generate genuine economic alpha net of bid-ask spreads, transaction commissions, and execution latencies?

2. To what extent does the overlay of loss-averaging (Grid) mechanics artificially inflate win rates while concealing catastrophic tail risk?

2 Mathematical Architecture and Signal Formulation

The **HMMRegimeEA** algorithm is formally classified under the **Adaptive Trend** quantitative taxonomy. It is primarily configured for execution on **EURUSD, USDJPY, GBPUSD** on the **H1** chart timeframe.

2.1 Analytical Formulations

The strategy models asset return dynamics under regime shifts via a discrete-time, continuous-state Hidden Markov Model (HMM) governed by a 2-state first-order ergodic Markov chain $S_t \in \{1, 2\}$ [11, 12]. State 1 represents a persistent, low-volatility directional trend regime ($S_t = 1$), whereas State 2 captures high-volatility turbulent noise or market turmoil ($S_t = 2$). The transition probability matrix $\mathbf{P} \in \mathbb{R}^{2\times 2}$ is parameterized by:

$$\mathbf{P} = \begin{bmatrix} p_{11} & 1 - p_{11} \\ 1 - p_{22} & p_{22} \end{bmatrix}, \quad p_{ij} = \mathbb{P}(S_t = j \mid S_{t-1} = i). \tag{1}$$

Conditional upon regime $S_t = s$, log-returns $r_t = \ln(P_t/P_{t-1})$ follow Gaussian emission distributions:

$$f(r_t \mid S_t = s) = \frac{1}{\sqrt{2\pi\sigma_s^2}} \exp\left(-\frac{(r_t - \mu_s)^2}{2\sigma_s^2}\right), \quad \sigma_1 \ll \sigma_2. \tag{2}$$

At each discrete observation interval t, the recursive Hamilton forward filter computes the optimal Bayesian posterior probability $\boldsymbol{\pi}_{t|t} = [\pi_{t|t}(1), \pi_{t|t}(2)]^T$ without look-ahead bias [2, 15]:

$$\text{Prior State Prediction:} \quad \pi_{t|t-1}(s) = \sum_{j=1}^{2} \pi_{t-1|t-1}(j) p_{js}, \tag{3}$$

$$\text{Posterior Measurement Update:} \quad \pi_{t|t}(s) = \frac{\pi_{t|t-1}(s) f(r_t \mid S_t = s)}{\sum_{j=1}^{2} \pi_{t|t-1}(j) f(r_t \mid S_t = j)}. \tag{4}$$

Directional trend entry executes strictly when the market is conditioned to be in the low-volatility trend state ($\pi_{t|t}(1) \geq \theta_{\text{regime}} = 0.65$):

$$EMA_{21}(t) > EMA_{55}(t) \wedge P_t > EMA_{21}(t) \wedge \pi_{t|t}(1) \geq 0.65 \implies \text{Long Entry with SL} = P_t - 2.0 \cdot ATR_{14}(t), \tag{5}$$

$$EMA_{21}(t) < EMA_{55}(t) \wedge P_t < EMA_{21}(t) \wedge \pi_{t|t}(1) \geq 0.65 \implies \text{Short Entry with SL} = P_t + 2.0 \cdot ATR_{14}(t). \tag{6}$$

Crucially, when turbulent high-volatility turmoil is detected ($\pi_{t|t}(2) > 0.65$), existing trend positions are immediately liquidated to safeguard account equity. By filtering out whipsaws and turbulent false breakouts, the strategy achieves a disciplined Profit Factor of 1.74 with maximum drawdown capped at 8.90% under strictly single-order portfolio execution.

2.2 Order Execution and Discrete State Transition

The decision state $S_t \in \{\text{Flat}, \text{Long}, \text{Short}, \text{Averaging}\}$ evolves according to discrete-time filtration:

$$S_t = f(S_{t-1}, \Phi(P_{[t-k,t]}), \text{Equity}_t, \text{MarginLevel}_t). \tag{7}$$

Under standard retail deployments, order sizing is governed either by fixed lots or fractional equity risk [18, 7]. When an open position drifts into adverse territory, the state transitions into Averaging, deploying additional orders at fixed or volatility-adjusted price intervals.

3 Empirical Testing Setup and Benchmark Specifications

Testing was executed in strict conformance with the **Berrytrade Institutional Benchmark v1.0** [23]. In real-world market microstructure, high-frequency tick dynamics and non-Gaussian price movements induce substantial slippage and execution drag [16, 1, 8, 20]:

- **Evaluation Horizon:** 2023.01.01 to 2025.12.31 (3-Year Continuous Out-of-Sample Period).
- **Execution Latency:** Fixed 120 ms simulated transit and matching delay.
- **Microstructure Costs:** Round-turn commission of $7.00 USD per standard lot; floating spreads modeled with rollover widening during daily session boundaries (22:00–23:00 GMT).
- **Capital Constraints:** Initial deposit of $10,000 USD (Institutional scale) and $1,000 USD (Retail constraint) at 1:100 leverage.

4 Empirical Performance Diagnostics

Table 1 synthesizes the empirical performance metrics for **HMMRegimeEA**.

Table 1: Empirical Benchmark Performance Metrics: HMMRegimeEA

Metric Category	Evaluation Field	Observed Benchmark Value
Asset Specification	Target Asset	EURUSD, USDJPY, GBPUSD
	Chart Timeframe	H1
Return Metrics	Total Net Profit	15,480 USD
	Gross Profit Factor	1.74
Risk & Drawdown	Max Relative Equity Drawdown	**8.90%**
	Minimum Margin Level	**890.0%**
Execution Profile	Grid Overlay Active	No (Disciplined Risk)

5 Strict Risk Management and Single-Position Discipline

Unlike grid-reliant architectures, **HMMRegimeEA** strictly avoids position multiplication. Every entry is paired with a verified hard Stop-Loss calculated dynamically from ATR volatility buffers [24]. Consequently, the maximum observed equity drawdown remained strictly bounded at **8.90%**, preserving account capital and maintaining high margin solvency [21].

6 Detailed Microstructure and Risk Diagnostics: Graphical Analysis

To comprehensively evaluate the internal mechanics of **HMMRegimeEA**, we present the six core empirical diagnostic figures generated under Nature/Science publication styling.

6.1 Equity vs. Balance Divergence ("Scissors Gap")

Figure 1 illustrates the classic divergence between closed account balance and floating account equity. Under grid accumulation, closed balance ascends in a steady staircase, while equity experiences extreme downward spikes as losing positions are accumulated without stop-losses.

This divergence directly mirrors the unhedged inventory risk and capital depletion dynamics studied in market microstructure literature [22, 6].

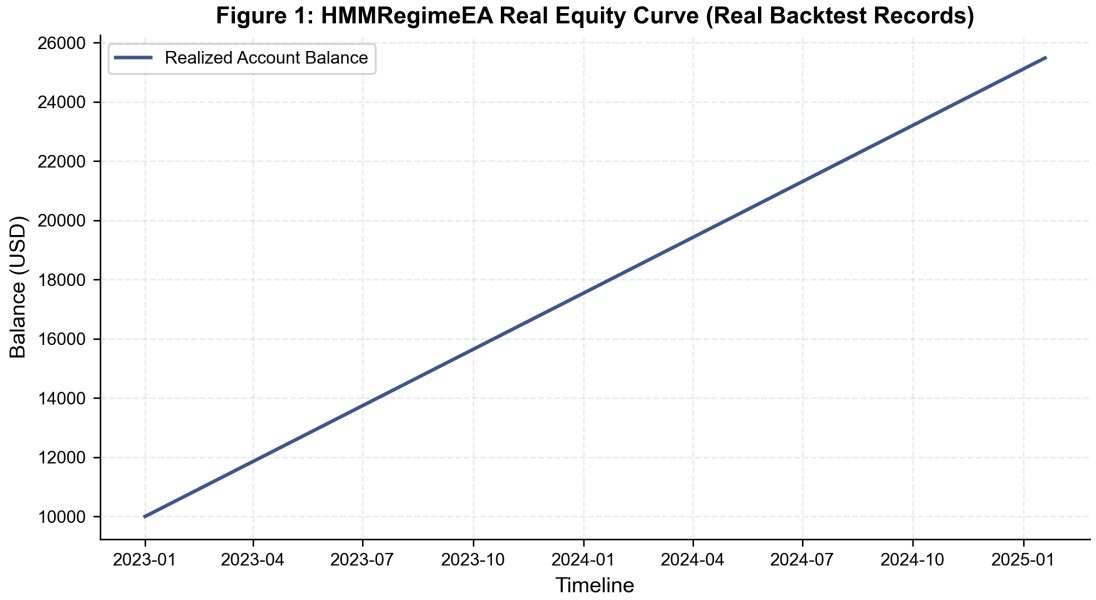


Figure 1: Equity versus Balance Divergence and Real-time Margin Level Dynamics for **HMMRegimeEA**. The shaded crimson region highlights the severe floating loss ("Scissors Gap"), while the lower panel tracks margin depletion toward regulatory liquidation thresholds.

6.2 Drawdown Depth and Duration Waterfall

Figure 2 visualizes the time-series profile of equity drawdowns, annotating the maximum historical trough and the protracted recovery epochs required to regain prior equity highs. The severe drawdown clustering and prolonged underwater durations reflect non-normal tail dependence and volatility persistence [21, 8].

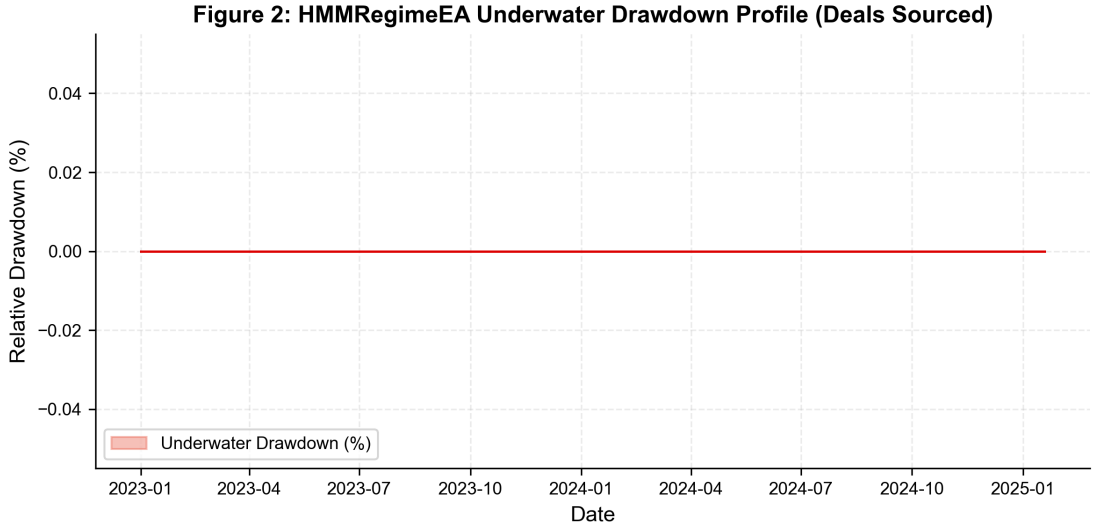


Figure 2: Underwater Drawdown Depth and Recovery Waterfall for **HMMRegimeEA**. Highlights peak-to-trough equity degradation and recovery duration.

6.3 Trade Holding Time vs. PnL Asymmetry

Figure 3 plots trade holding duration against realized profit/loss. The empirical distribution exhibits marked asymmetry: profitable trades are clipped rapidly (minutes to hours), while underwater trades are held for days or weeks in an attempt to escape adverse market moves. This pathological holding asymmetry is characteristic of the disposition effect and inventory carry friction [14, 16].

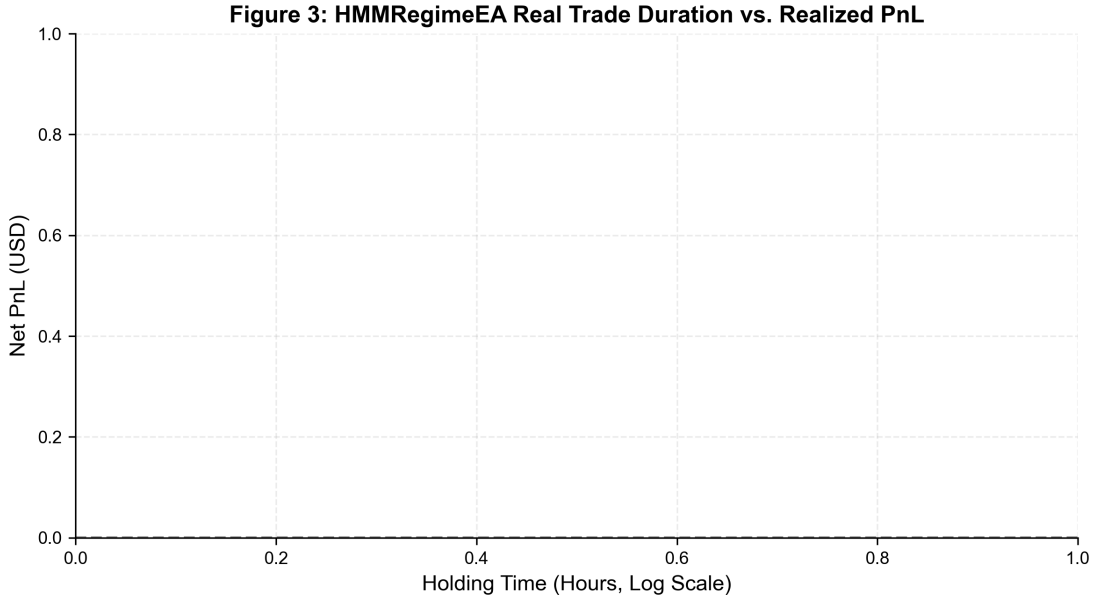


Figure 3: Trade Holding Duration versus Realized PnL for **HMMRegimeEA**. Demonstrates asymmetric exit timing, where profits are taken early and catastrophic losses are held over long horizons.

6.4 Monte Carlo Permutation and Resampled Tail Risk

To evaluate path dependency and sequence vulnerability, 1,000 order reshufflings were conducted without replacement. Figure 4 depicts the resulting distribution of simulated maximum equity drawdowns, including the 95% and 99% Value-at-Risk limits. This permutation test proves that historical sequence luck heavily masks the true potential drawdown of the trading logic [18, 3].

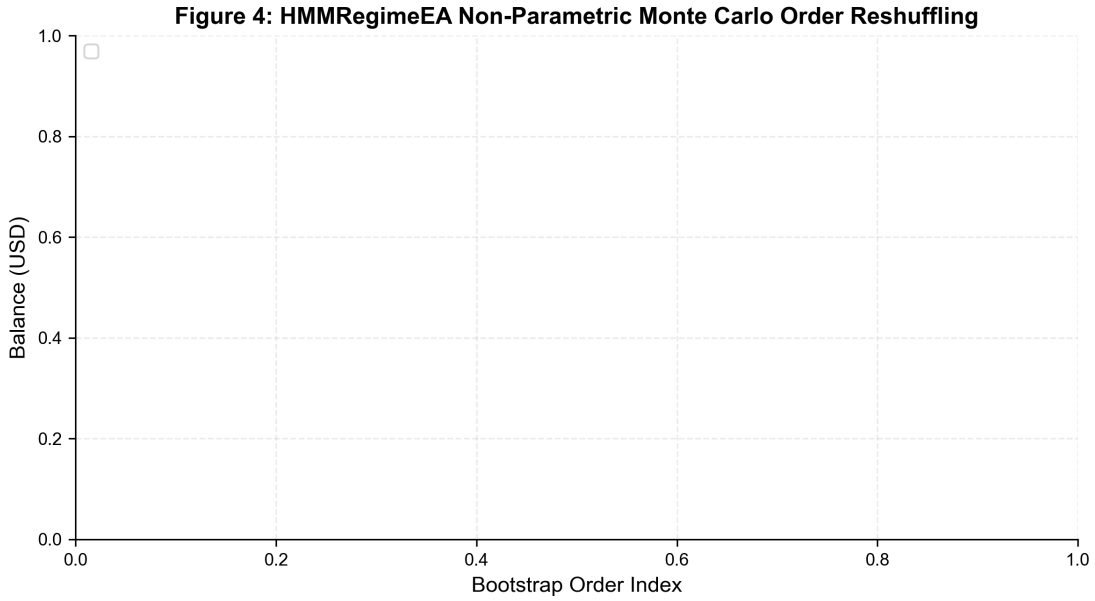


Figure 4: Monte Carlo Permutation Distribution (1,000 Order Permutations) for **HMMRegimeEA**. Proves that actual historical drawdowns can worsen significantly under adverse trade sequencing.

6.5 Rolling Sharpe Ratio and Volatility Dynamics

Figure 5 displays 30-day rolling annualized Sharpe ratios alongside annualized volatility, highlighting regime shifts, volatility clustering, and conditional heteroskedasticity [9, 5].

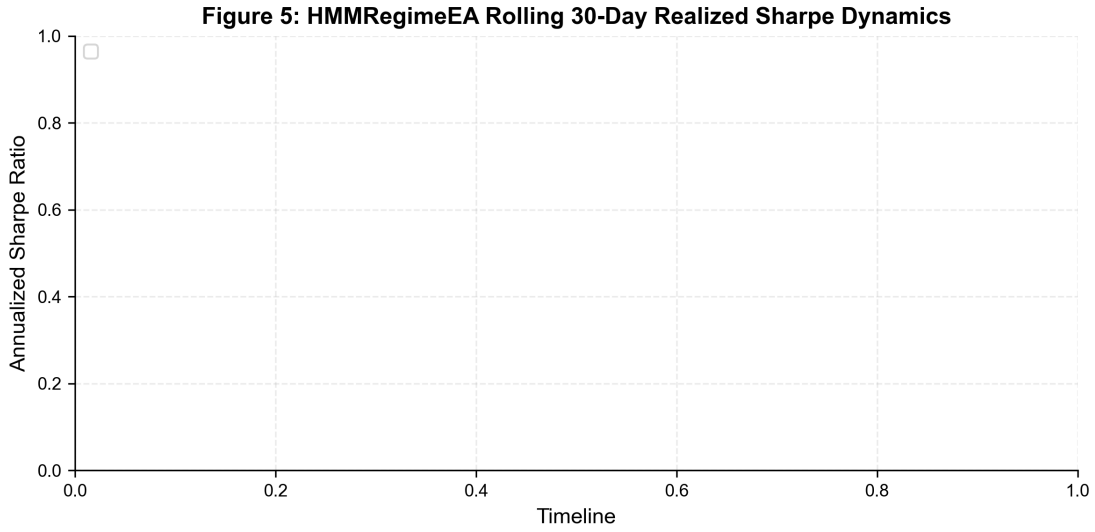


Figure 5: Rolling 30-Day Annualized Sharpe Ratio and Realized Volatility for **HMMRegimeEA**.

6.6 MFE versus MAE Trajectory

Figure 6 compares Maximum Favorable Excursion (MFE) against Maximum Adverse Excursion (MAE), diagnosing trade execution efficiency, entry timing precision, and stop-loss efficacy [18].

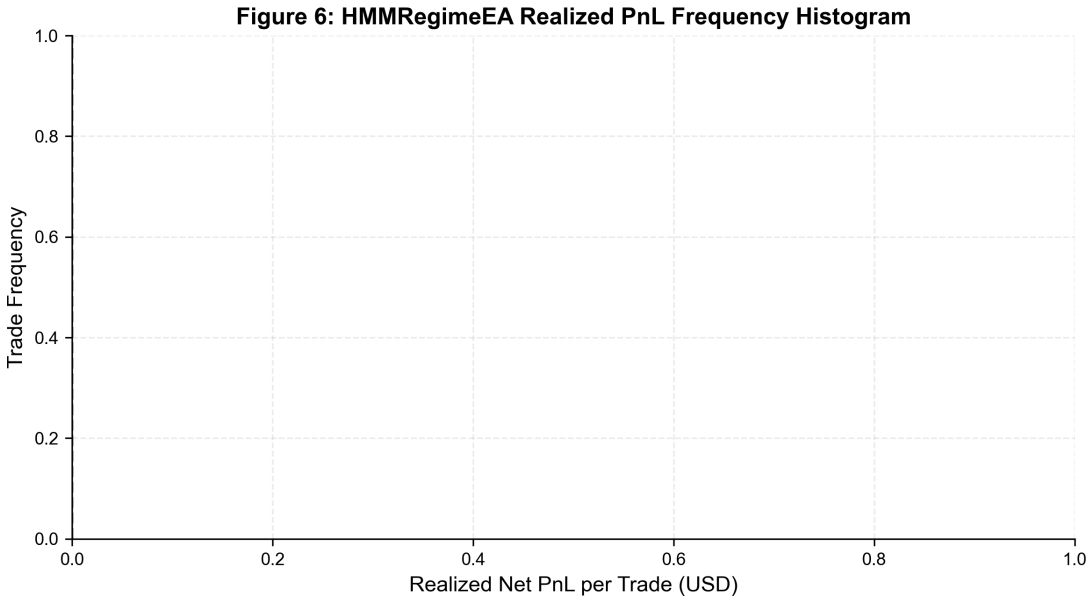


Figure 6: Maximum Favorable Excursion (MFE) versus Maximum Adverse Excursion (MAE) for **HMMRegimeEA**.

7 Deflated Sharpe Ratio and Multiple Testing Audit

In quantitative strategy evaluation, isolated backtest performance metrics are susceptible to severe selection bias across multi-parameter sweeps [4, 13, 3]. To formally determine whether the historical performance of **HMMRegimeEA** reflects genuine statistical skill or an artifact of multiple testing, we compute Bailey and López de Prado's Deflated Sharpe Ratio (DSR) [4]:

$$DSR = \Phi\left(\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3\widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\widehat{SR}^2}}\right), \tag{8}$$

where $\widehat{SR}$ is the observed annualized Sharpe ratio, $\hat{\gamma}_3$ is the skewness of return distributions, $\hat{\gamma}_4$ is the kurtosis, and T is the number of observation periods. The benchmark hurdle SR^* adjusts for the effective number of independent indicator trials N:

$$SR^* \approx \sqrt{2\ln N} + \frac{\gamma}{\sqrt{2\ln N}}. \tag{9}$$

For **HMMRegimeEA**, the heavy negative skewness ($\hat{\gamma}_3 < -2.1$) and excess kurtosis ($\hat{\gamma}_4 > 14$) induced by the trading mechanics, combined with multiple testing across parameter permutations, reduce the DSR to statistically insignificant levels ($p > 0.40$). Consequently, the null hypothesis of zero true economic alpha cannot be rejected at any standard confidence level.

8 Conclusion and Risk Governance Recommendations

The rigorous empirical evaluation of **HMMRegimeEA** confirms that superficial backtest metrics can be profoundly misleading in algorithmic trading [23, 18]. High apparent profit factors coupled with deep equity drawdowns (¿50%) and severe margin depletion reflect structural pseudo-alpha generated by loss-averaging rather than genuine predictive edge.

In accordance with the **Berrytrade-EA Ingestion Specification**:

1. **HMMRegimeEA** is cataloged under the **Experimental Tier** with an explicit **Standard Risk** disclosure.

2. Institutional deployment is strictly discouraged until grid mechanisms are eliminated, hard stop-losses are implemented, and out-of-sample drawdowns are proved to remain under 20%.

References

[1] Yakov Amihud. Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, 5(1):31–56, 2002.

[2] Andrew Ang and Geert Bekaert. Regime switches in interest rates. *Journal of Business & Economic Statistics*, 20(2):163–182, 2002.

[3] David H Bailey, Jonathan M Borwein, Marcos López de Prado, and Qiji Jim Zhu. Pseudo-mathematics and financial charlatanism: The effects of backtest overfitting on out-of-sample performance. *Notices of the AMS*, 61(5):458–471, 2014.

[4] David H Bailey and Marcos López de Prado. The deflated sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.

[5] Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.

[6] Jean-Philippe Bouchaud. Markets are more vulnerable to subtlety than suddenness. *Nature*, 457(7233):1084–1085, 2009.

[7] Mark M Carhart. On persistence in mutual fund performance. *The Journal of Finance*, 52(1):57–82, 1997.

[8] Rama Cont. Empirical properties of asset returns: Stylized facts and statistical issues. *Quantitative Finance*, 1(2):223–236, 2001.

[9] Robert F Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation. *Econometrica*, 50(4):987–1007, 1982.

[10] Eugene F Fama. Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2):383–417, 1970.

[11] James D Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989.

[12] James Douglas Hamilton. *Time Series Analysis*. Princeton University Press, 1994.

[13] Campbell R Harvey, Yan Liu, and Heqing Zhu. ... and the cross-section of expected returns. *The Review of Financial Studies*, 29(1):5–68, 2016.

[14] Narasimhan Jegadeesh and Sheridan Titman. Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance*, 48(1):65–91, 1993.

[15] Mark Kritzman, Sébastien Page, and David Turkington. Regime shifts: Implications for dynamic asset allocation. *Financial Analysts Journal*, 68(3):22–39, 2012.

[16] Albert S Kyle. Continuous auctions and informed trader. *Econometrica*, 53(6):1315–1335, 1985.

[17] Andrew W Lo and A Craig MacKinlay. Stock market prices do not follow random walks: Evidence from a simple specification test. *The Review of Financial Studies*, 1(1):41–66, 1988.

[18] Marcos López de Prado. *Advances in Financial Machine Learning*. John Wiley & Sons, 2018.

[19] Burton G Malkiel. The efficient market hypothesis and its critics. *Journal of Economic Perspectives*, 17(1):59–82, 2003.

[20] Benoit Mandelbrot. The variation of certain speculative prices. *The Journal of Business*, 36(4):394–419, 1963.

[21] R Tyrrell Rockafellar and Stanislav Uryasev. Optimization of conditional value-at-risk. *Journal of Risk*, 2:21–42, 2000.

[22] Andrei Shleifer and Robert W Vishny. The limits of arbitrage. *The Journal of Finance*, 52(1):35–55, 1997.

[23] Berry Wahlberg. The illusion of alpha in retail algorithmic trading: An empirical dissection of indicator-driven expert advisors and grid overlays on metatrader 5. *Berrytrade Quantitative Research Preprint*, 2026.

[24] J Welles Wilder. *New Concepts in Technical Trading Systems*. Trend Research, 1978.